Trajectory Prediction of Marine Drifting Objects

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Abstract

Predicting the trajectories of marine drifting objects is critical for ocean monitoring, pollution tracking, and maritime search and rescue. Using the NOAA Global Drifter Program 6-hour dataset, we formulate a 48-hour-to-24-hour displacement prediction task and compare physics-based baselines, classical machine learning, and deep learning models. A simple advection baseline achieves 8.72 km mean geodesic error; incorporating a physics-residual formulation progressively reduces error through Ridge (7.80 km), Random Forest (7.25 km), and Physics-Constrained RF (6.86 km). A two-layer LSTM with advection residual learning reaches the best overall accuracy of **5.88 km**, outperforming a Transformer-ODE variant (7.67 km) on this dataset. Our results show that a compact recurrent architecture with a physics prior is sufficient and effective for this problem.

1 Introduction

Predicting the trajectories of marine drifting objects, such as ocean buoys, floating debris, and oil spill particles, is important for oceanography, environmental monitoring, and maritime search and rescue. Accurate drift prediction can help track marine pollution, support oil spill response, analyze ocean surface currents, and narrow down search regions after accidents.

A real-world motivation is the search for floating debris after aviation or shipping incidents. For example, after the disappearance of Malaysia Airlines Flight MH370, ocean drift modeling was used to estimate possible debris pathways and support search area analysis. Such cases show that reliable prediction of marine drifting motion can have significant practical value.

However, marine drifting trajectories are affected by complex factors such as ocean currents, wind, waves, and local environmental conditions. Simple physics-based models are interpretable but may be limited when the dynamics are nonlinear or when environmental data are noisy. In this project, we aim to use the NOAA Global Drifter Program 6-hour dataset[1] to predict future positions of drifting objects and compare physics-based baselines with machine learning and deep learning models.

2 Related Work

Prior work related to this project can be grouped into three main directions: physics-based drift modeling, Lagrangian particle-tracking frameworks, and data-driven trajectory prediction.

Physics-based drift modeling. Traditional marine drift prediction often models floating objects as Lagrangian particles transported by ocean currents, wind, and wave-induced effects. This approach has been used in real search and rescue scenarios, such as studies of possible MH370 debris drift paths [1, 2]. These models are interpretable, but their accuracy can be sensitive to uncertain environmental forcing, initial conditions, and object-specific drift behavior.

[1] https://www.ncei.noaa.gov/access/metadata/landing-page/bin/iso?id=gov.noaa.nodc:AOML-GDP-6hr

Particle-tracking frameworks. Numerical frameworks such as Parcels simulate ocean particle transport using gridded ocean circulation data [3]. They are useful for modeling passive particles such as plastics, biological material, or floating debris, but they depend strongly on the quality and resolution of the underlying ocean current fields. Our project instead focuses on learning motion patterns directly from observed drifter trajectories.

Data-driven trajectory prediction. Recent studies have applied machine learning and deep learning models, including LSTM, CNN-LSTM, ConvLSTM, and Transformer-based methods, to ocean trajectory prediction [4, 5]. These methods can capture nonlinear temporal patterns, but they may require sufficient training data and can be less interpretable than physics-based models.

Our project is situated between these directions. Using NOAA Global Drifter Program observations [6], we will compare a physics-informed advection baseline, classical machine learning models, and sequence-based deep learning models under the same trajectory prediction setting.

3 Method / Approach

3.1 Problem Formulation

We construct supervised samples via a sliding window of $T = 8$ steps (48-hour history):

$$X_t = \big\{(\mathrm{lat}_{t-k},\, \mathrm{lon}_{t-k},\, u_{t-k},\, v_{t-k},\, \mathbf{f}_{t-k})\big\}_{k=T-1}^{0}, \tag{1}$$

where (u_t, v_t) are zonal/meridional velocities and $\mathbf{f}_t$ auxiliary features. The prediction target is the 24-hour displacement ($\Delta = 4$ steps):

$$\Delta Y_t = \big(\mathrm{lat}_{t+\Delta} - \mathrm{lat}_t,\ \mathrm{lon}_{t+\Delta} - \mathrm{lon}_t\big). \tag{2}$$

Models are evaluated by MAE and RMSE on $[\Delta\mathrm{lat}, \Delta\mathrm{lon}]$, and mean geodesic error (km) as the primary metric:

$$d_{\mathrm{geo}}(\hat{y}, y) = 2R \arcsin\sqrt{\sin^2 \tfrac{\Delta\phi}{2} + \cos\phi_1 \cos\phi_2 \sin^2 \tfrac{\Delta\lambda}{2}}, \quad R = 6371\,\mathrm{km}. \tag{3}$$

3.2 Baseline Methods

We build a three-tier baseline to isolate the contribution of each modeling layer.

Tier 1: Physics. *Persistence*: $\hat{x}_{t+\Delta} = x_t$. *Advection*: project the last observed velocity over $\Delta t = 24\,\mathrm{h}$,

$$\hat{x}_{t+\Delta} = x_t + v_t\,\Delta t. \tag{4}$$

The in-situ velocity v_t already integrates all forces on the drifter, so Eq. (4) is the strongest zero-parameter physical prediction available; no additional corrections are applied.

Tier 2: Classical ML. X_t is flattened (168 dims) and augmented with 7 physics-summary features derived from Eq. (4) (advection prediction, window-mean advection, velocity tendency, speed variance). Two models are trained:

$$\text{Ridge:} \quad \hat{y} = Xw, \quad w^* = \arg\min_w \|Xw - y\|^2 + \alpha\|w\|^2, \tag{5}$$

$$\text{Random Forest:} \quad \hat{y} = \frac{1}{B}\sum_{b=1}^{B} f_b(X). \tag{6}$$

Tier 3: Physics-constrained RF (PC-RF). We adopt a residual formulation that separates physics from data-driven correction:

$$\hat{y} = \underbrace{v_t\,\Delta t}_{\text{advection}} + \underbrace{f_\theta(X_t)}_{\text{residual correction}}, \tag{7}$$

where f_θ is a Random Forest trained on residuals $y - v_t\Delta t$. The advection term is excluded from the input of f_θ, so the physics prior enters only through the target, keeping the two contributions cleanly separated. This formulation makes the physics contribution quantifiable and provides a structurally motivated starting point for the sequence models described below.

3.3 LSTM

We treat the LSTM as the deep-learning version of the PC-RF baseline. The network reads the 48-hour window X_t and predicts the 24-hour displacement ΔY_t. A two-layer LSTM with hidden size 128 encodes the sequence, and we keep the hidden state at the final step as a summary of the window. A small MLP head with shape $128 \to 64 \to 2$ turns this state into the two outputs $[\Delta\mathrm{lat}, \Delta\mathrm{lon}]$. We standardize each input feature with statistics from the training split, so the validation and test windows stay out of the fitting step. Figure 1 shows the full model.

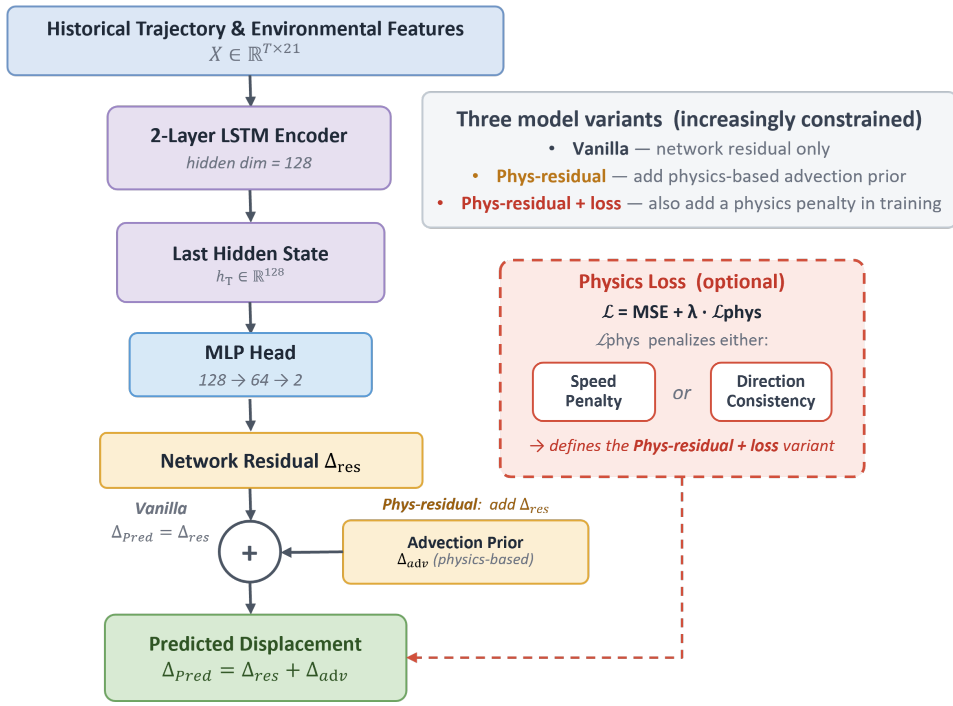

Figure 1: LSTM architecture. A two-layer LSTM encodes the 48-hour window, an MLP head predicts a residual, and the advection prior is added back to form the final displacement. The three variants differ only in where physics enters.

We study three variants that differ in where physics enters. The vanilla model predicts the displacement directly from the network output. The physics-residual model reuses the advection prior from Eq. (7) and learns only the correction on top of it,

$$\hat{y} = \underbrace{v_t\,\Delta t}_{\text{advection}} + f_\theta(X_t), \tag{8}$$

which copies the PC-RF structure and lets us check whether the residual idea that helped the Random Forest also helps a sequence model. The third variant keeps this structure and adds a physics term to the training loss,

$$\mathcal{L} = \mathcal{L}_{\mathrm{MSE}} + \lambda\,\mathcal{L}_{\mathrm{phys}}, \tag{9}$$

where $\mathcal{L}_{\mathrm{phys}}$ penalizes predicted drift speeds above a physical limit of 2 m/s. We also ran a direction-consistency penalty that pushes the predicted displacement toward the advection direction.

We pick the model on validation mean geodesic error instead of the training loss, so selection matches the metric we report. Training runs Adam with a cosine schedule from 10^{-3} to 10^{-5}, batch size 256, and early stopping with patience 7. We repeat every variant over three seeds and report the mean together with the spread across seeds.

3.4 Transformer

Overall Framework: The forward process is:

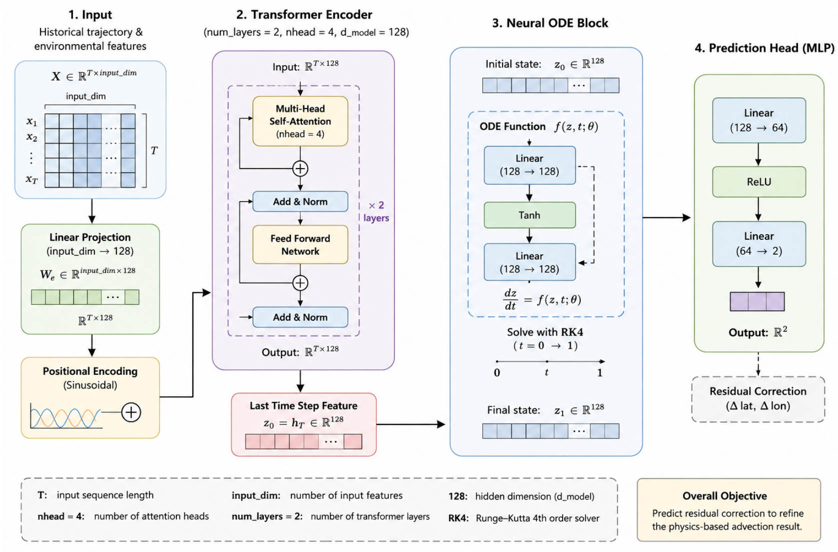

Figure 2: Model Architecture

$$X \to \text{Transformer Encoder} \to z_0 \to \text{Neural ODE} \to z_T \to \text{MLP Head} \to \hat{r} \tag{10}$$

Final prediction:

$$\hat{y} = \Delta x_{adv} + \hat{r} \tag{11}$$

where: $\hat{r}$ is the predicted residual correction.

Loss Function

Prediction Loss: The training objective primarily minimizes residual prediction error:

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^{N} ||\hat{r}_i - r_i||^2 \tag{12}$$

The implementation additionally uses Smooth L1 loss during validation for robustness.

Physics Regularization: To stabilize residual magnitude, an auxiliary physics regularization term is added:

$$\mathcal{L}_{phys} = \frac{1}{N} \sum_{i=1}^{N} ||\hat{r}_i||^2 \tag{13}$$

The final objective becomes:

$$\mathcal{L} = \mathcal{L}_{MSE} + \lambda \mathcal{L}_{phys} \tag{14}$$

where:$\lambda = 0.01$ controls the regularization strength. **Training Strategy**: The framework uses AdamW optimization: learning rate: 10^{-4}, weight decay: 10^{-5}.AdamW improves generalization by decoupling weight decay from adaptive optimization.

A ReduceLROnPlateau scheduler is used: factor: 0.5, patience: 3 epochs.This strategy reduces learning rate when validation performance stagnates.

4 Results & Analysis

4.1 Dataset and Experimental Setup

We use the NOAA Global Drifter Program (GDP) 6-hour interpolated dataset [6] covering 2022–2025. Continuous trajectories are segmented into supervised samples via a sliding window of $T=8$ steps (48-hour history), predicting the latitude/longitude displacement ΔY after $\Delta=4$ steps (24 hours), as defined in Section 3.1. Each sample has input shape $\mathbb{R}^{8\times 21}$ and target shape $\mathbb{R}^2$.

The 21 features per time step include: four core trajectory variables (`lat`, `lon`, `ve`, `vn`); four optional sensor variables (`temp`, error estimates); nine derived physical features (speed, direction, stepwise displacements, velocity changes, accelerations, step distance); and four time encodings (day-of-year and hour-of-day sine/cosine). Standardization statistics are fitted on the training split only to avoid data leakage.

All four years are concatenated and split 70/15/15, giving 727,755 training, 153,379 validation, and 149,158 test windows. All models share the same held-out test set. Table 1 summarizes final performance across all models.

Table 1: All models on the test set (149,158 samples), ranked by mean geodesic error.

Model	Mean geo (km)	Median geo (km)
Persistence (Tier 1)	19.108	14.208
Advection (Tier 1)	8.721	6.610
Transformer (vanilla)	8.920	5.390
Transformer (multistep)	8.320	5.070
Ridge (Tier 2)	7.801	5.993
Transformer (phys-residual)	7.670	5.230
Random Forest (Tier 2)	7.251	5.488
PC-RF (Tier 3)	6.859	5.147
LSTM (vanilla)	5.962	4.330
LSTM (phys-residual+loss)	5.885	4.276
LSTM (phys-residual)	**5.880**	4.280

4.2 Baseline Results

We evaluate five baselines on the held-out test set (Table 2). Ridge regularisation strength $\alpha=1000$ is selected by a validation sweep over $\{10^{-2}, \ldots, 10^4\}$; RF and PC-RF use 200 trees, max depth 14, and `min_samples_leaf` $= 10$.

Table 2: Baseline model performance on the test set (149,158 samples).

Model	MAE lat	MAE lon	RMSE lat	RMSE lon	Mean geo (km)	Median geo (km)
Persistence (Tier 1)	0.09797	0.16401	0.14693	0.79057	19.108	14.208
Advection (Tier 1)	0.05097	0.07267	0.08019	0.65753	8.721	6.610
Ridge (Tier 2)	0.04564	0.06506	0.07053	0.57196	7.801	5.993
Random Forest (Tier 2)	0.04281	0.06178	0.06550	0.70356	7.251	5.488
PC-RF (Tier 3)	0.04022	0.05906	0.06256	0.64392	**6.859**	**5.147**

Advection alone reduces error from 19.1 km to 8.7 km, confirming in-situ velocity as a strong prior. Classical ML (Ridge: 7.8 km, RF: 7.3 km) captures additional nonlinear patterns from the 48-hour history. PC-RF achieves the best baseline at 6.86 km by restricting the RF to learn only the structured residual $y - v_t\Delta t$ (Eq. (7)).

Figure 3 reveals two patterns. At high speeds (>0.36 m/s), ML models converge toward the advection baseline, indicating that instantaneous physical forcing dominates and historical context becomes less useful. Across latitude bands, ML models remain robust where persistence degrades: error peaks in the tropics (Ekman currents) and at mid-latitudes (mesoscale eddies) are largely absent in the ML models. Both patterns suggest the remaining residuals are governed by high-frequency variability, motivating sequence-based architectures.

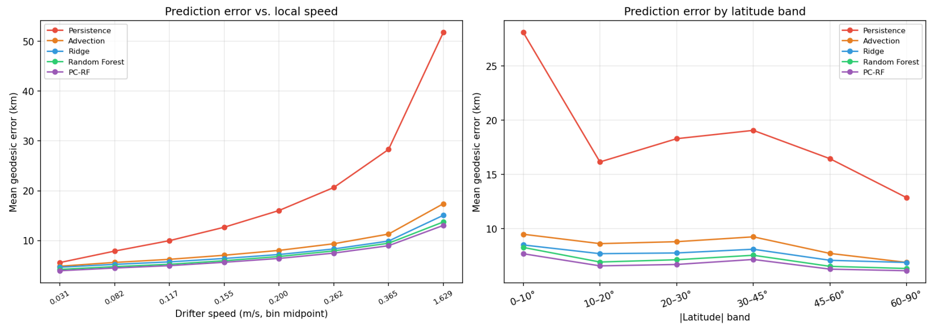

Figure 3: Mean geodesic error stratified by local drifter speed (left) and absolute latitude band (right), shown for all baseline models.

4.3 LSTM Results

We reuse the baseline data pipeline; the test split matches exactly so numbers are directly comparable. Table 3 reports the three variants, each averaged over seeds 7, 42, and 123.

Table 3: LSTM variants on the test set, averaged over three seeds. ± denotes best-to-worst seed gap.

Model	MAE lat	MAE lon	RMSE lat	RMSE lon	Mean geo (km)	Median geo (km)
LSTM (vanilla)	0.03470	0.05475	0.05500	0.71845	5.962±0.038	4.330
LSTM (phys-residual)	0.03429	0.05254	0.05435	0.63739	**5.880**±0.024	4.280
LSTM (phys-residual+loss)	0.03427	0.05273	0.05443	0.63639	5.885±0.010	4.276

All three variants beat PC-RF (6.86 km). The seed range sits near 0.03 km, which sets a floor for reading small gaps. The advection prior gives a real gain (5.96 → 5.88 km), as this gap exceeds the noise floor. Adding the speed penalty changes the error by only 0.005 km, within seed variance, so the physics loss adds nothing. The direction-consistency penalty actively hurt performance (best result 6.81 km), because it pulls predictions toward the advection direction even when the true motion turns away from it, confirming that physics helps through residual structure, not through the training objective.

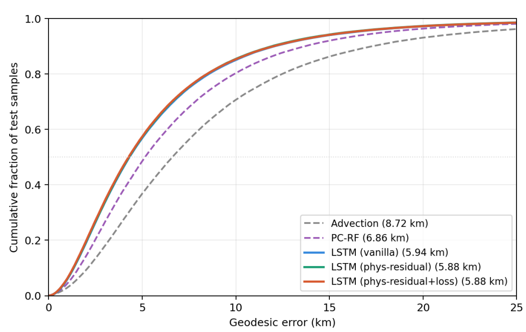

Figure 4: Cumulative distribution of per-sample geodesic error on the test set. The three LSTM variants overlap and stay left of PC-RF and advection.

Figure 4 shows the gain holds across the full error distribution, not only at the mean.

Ablation study. We vary one factor at a time around the reference setting and select on validation mean geodesic error (Table 4). The seed row spans about 0.04 km, which marks the noise floor. Hidden size 64 falls clearly behind 128, while 128 and 256 tie, so 128 is enough. A single layer or no dropout costs a little, while the physics-loss weight makes no difference at any value, which repeats the earlier point that the physics loss adds nothing. The core-4 input hurts the most and raises the error by about 0.18 km, so the engineered features carry signal beyond raw position and velocity.

Table 4: One-factor-at-a-time ablation, validation mean geodesic error (km). Each row varies one factor around the reference configuration (hidden 128, 2 layers, dropout 0.1, $\lambda = 1.0$, all 21 features, seed 42), which scores 5.63.

Factor	Settings	Val mean geo (km)
Hidden size (2 layers)	64 / 128 / 256	5.79 / 5.63 / 5.63
Layers (hidden 128)	1 / 2 / 3	5.75 / 5.63 / 5.67
Dropout	0.0 / 0.1 / 0.2	5.69 / 5.63 / 5.64
Physics loss λ	0 / 0.01 / 0.1 / 1.0	5.65 / 5.65 / 5.64 / 5.63
Features	core-4 / all-21	5.81 / 5.63
Seed	7 / 42 / 123	5.67 / 5.63 / 5.65

4.4 Transformer Results

Table 5: Transformer variants on the test set.

Model	MAE lat	MAE lon	RMSE lat	RMSE lon	Mean geo (km)	Median geo (km)
Transformer (vanilla)	0.03920	0.05475	0.05500	0.7724	8.92	5.39
Transformer (phys-residual)	0.03910	0.05254	0.05910	0.7425	**7.67**	5.23
Transformer (multistep)	0.05427	0.03025	0.05443	0.7569	8.32	5.07

Table 5 shows that the physics-residual formulation reduces mean geodesic error from 8.92 to 7.67 km, consistent with the pattern seen in baselines and LSTM. The multistep variant achieves the best median (5.07 km) at the cost of a higher mean, suggesting it reduces typical errors but is more sensitive to outliers. Across all variants, median geodesic error is substantially lower than mean error, indicating that large-error outliers are relatively infrequent but drive the mean upward.

Despite stable convergence (Figure 5), the Transformer variants underperform all ML tiers and the LSTM. Three factors explain this gap. First, the sequences are short (T=8) and exhibit strong temporal continuity; the recurrent inductive bias of LSTM is well matched to this structure, while self-attention provides limited benefit without longer-range context. Second, the dataset is relatively small for a Transformer, which typically requires more data to learn meaningful attention patterns than a compact LSTM. Third, ocean drift dynamics are smooth and slowly varying, aligning naturally with the LSTM memory-cell structure rather than global attention.

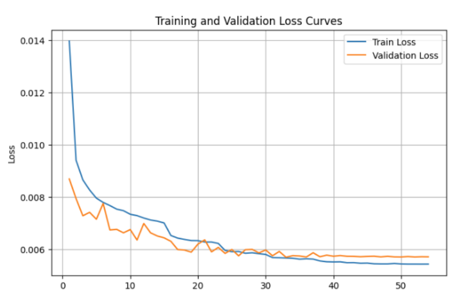

Figure 5: Training and validation loss curves of the Transformer-ODE model. Both losses converge after ~30 epochs with no severe overfitting.

Longitude Error Analysis: Longitude RMSE exceeds latitude RMSE across all Transformer variants, consistent with the baseline and LSTM findings. This asymmetry reflects east-west current variability, mesoscale eddy activity, and the $\cos(\phi)$ coordinate scaling factor that amplifies longitude sensitivity at higher latitudes.

5 Conclusion

This project systematically compared physics-based, classical ML, and deep learning approaches for 24-hour marine drifter trajectory prediction on the NOAA GDP dataset, yielding three main takeaways.

Physics prior explains most of the drift, while residual learning adds further gains. Advection alone reduces error from 19.1 km to 8.7 km, confirming in-situ velocity as a strong prior. Residual learning — decomposing prediction into an advection term plus a learned correction — is the single most effective design choice, improving performance at every tier from PC-RF to LSTM, with each correction adding a smaller further gain on top of the physics prior. Notably, encoding physics through the residual structure is sufficient; adding an explicit physics penalty to the training loss provides no further gain.

A compact recurrent model is the right architecture for this problem. The LSTM (phys-residual) achieves the best overall accuracy at 5.88 km, outperforming the Transformer-ODE (7.67 km). On the short sequences and smooth drift dynamics of this dataset, the recurrent inductive bias proves more effective than global self-attention.

Longitude error persistently exceeds latitude error. This asymmetry is consistent across all models and reflects east-west current variability, mesoscale eddy activity, and the $\cos(\phi)$ coordinate scaling at higher latitudes — a structural challenge that neither architecture resolves.

Limitations include the fixed 24-hour prediction horizon and reliance on a single dataset. Future work could explore multi-step prediction, incorporation of external environmental forcing such as wind fields and sea surface height, and larger pretraining corpora to better leverage attention-based architectures.

6 Code & GitHub

Code is publicly available at https://github.com/JaneeetHe/Trajectory-Prediction-of-Marine-Drifting-Objects.

7 Team Member Contribution

Jingyi He completed the data processing and early-stage project setup, including raw NOAA drifter data collection, trajectory cleaning, supervised window construction, feature engineering, and processed dataset preparation. Additional contributions included final slides preparation and delivery of the full oral presentation.

Danni Ma designed and implemented the baseline models (persistence, advection, Ridge, Random Forest, and PC-RF), conducted experiments and analysis, wrote the corresponding report sections and slides, and integrated contributions from all team members into the final report.

Yiting Wang built the LSTM module (physics-residual architecture, three variants, multi-seed training, and ablation study) and made the LSTM figures, slides, and report sections.

Ruixin Fu has researched about the reasonable Transformer architecture for this subject, implemented the model experiments, and done several ablation studies, also made figures and slides of Transformer parts for final Presentation.

Bonus: Course Evaluation Submission

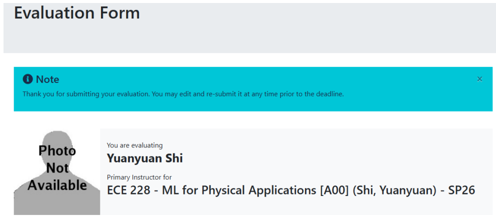

Figure 6: *
(a) Yiting Wang

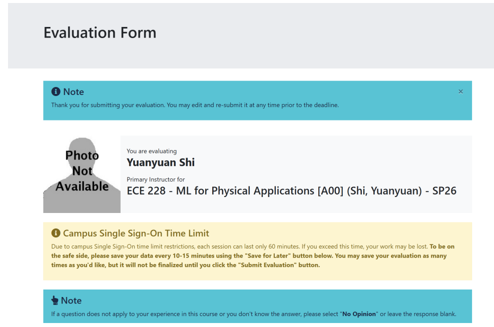

Figure 7: *
(b) Jingyi He

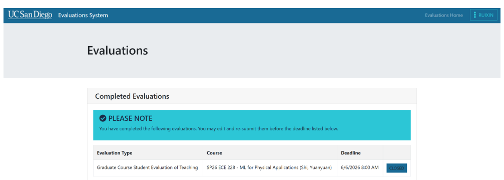

Figure 8: *
(c) Ruixin Fu

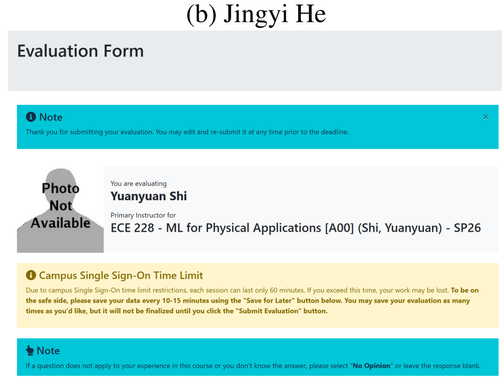

Figure 9: *
(d) Danni Ma

Figure 10: Course evaluation submission confirmations.

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